

Feature Name	Previous or Created	Time Detrending	Intuitive Meaning	How it's calculated
Spectral Entropy	Previous		How spread out the signal's energy is across frequencies (high = more spread out)	Calc power spectral density with Welch's method, normalize it, find Shannon entropy
Spectral Centroid	Previous		Where the average frequency of the signal lies (low frequencies = low centroid)	Weighted average of frequencies
Spectral Flatness	Previous	Yes	Tells whether energy is concentrated in a few frequencies or spread evenly across many frequencies. High spectral flatness = more spread out.	Calc PSD, find geometric mean of PSD, divide it by arithmetic mean of PSD
Spectral Bandwidth	Previous		How wide the signal's energy is spread around its average frequency. Like standard deviation. Large bandwidth = frequencies more spread out from centroid.	Average squared distance from centroid, weighted by power.
Std_dev_time_diff	Previous		Std deviation of the inter peak times	Find the peaks, find time differences between the peaks. Find std. deviation of those differences.
Mean_time_diff	Previous	Yes	Average time interval between peaks (mean of inter peak times)	Find the peaks, find time differences between the peaks. Find mean of those differences.
Median_time_diff	Previous	Yes	Median of the time differences (median of inter peak times)	Find the peaks, find time differences between the peaks. Find median of those differences.
mean_peak_magnitude	Previous		It represents the typical strength of the vibration spikes caused by	Average magnitude of the detected peaks in a signal
max_peak_magnitude	Previous		It captures the strongest single vibration event recorded by the ac	Find peaks, take largest of those peaks
std_peak_magnitude	Previous		This measures how much the detected peak sizes vary from one	Find standard deviation of peak magnitudes
peaks_per_second	Previous		measures the rate of detected peaks in the signal	It is calculated by dividing the number of detected peaks by the t
percent_time_above_threshold	Previous		measures the proportion of the signal that remains above a peak-based activity threshold	It is calculated by finding the fraction of signal values greater than the smallest detected peak magnitude, which gives an estimate of how often the signal stays in an active vibration range.
num_boilings	Previous	Yes	Number of distinct patterns of boiling found	From previous team's code
unused_peak_proportion	Previous		Fraction of peaks not found to be part of any detected regime	Count peaks not assigned to any regime, divide by total peaks (regime assignments decided by previous team's code in num boilings)
skewness	Created		Assymetry of the signal's distribution around its mean (positive skew = occasional high spikes)	The third standardized central moment
kurtosis	Created		Measures how heavy the tails of the distribution are. High kurtosis = occasional very high peaks	The fourth standardized central moment
crest factor	Created		How extreme the peaks are relative to the overall "energy" of the signal. High crest factor = short, sharp spikes	Peak amplitude divided by root mean square of the signal
Burstiness	Created		Measures how irregular the signal is. High burstiness = periods of rapid activity followed by slow periods	std deviation of inter peak arrivals divided by mean of inter peak arrivals
spectral_flux	Created		How quickly the spectrum of a signal changes over time	Flux is the sum of the squared differences of the spectrogram across consecutive time periods. Find differences across time periods within each frequency, then square the differences, sum up to find total in each time period, find average of that.
low_band_power	Created		The fraction of total signal energy in low frequency range	Take the sum of the power spectral density, then find the sum of the PSD that lives within each defined band. Then normalize each and return the proportion of each band power. Low band power = 0-200 Hz, Med Band Power = 200 - 400 Hz, High 400 - 500 Hz
mid_band_power	Created		The fraction of total signal energy in medium frequency range	
high_band_power	Created		The fraction of total signal energy in high frequency range	

Feature Name	Previous or Created	Time Detrending	Intuitive Meaning	How it's calculated
high_to_low_ratio	Created	Yes	Ratio of high band power to lower band power	Take high band power and divide it by lower band power
mean_spectral_entropy_time	Created		How spread out frequency is, averaged across time	Calc the spectrogram. Then normalize each time slice. Now calculate entropy at each time period. Take the mean of the entropy values across the different time periods
var_spectral_centroid_time	Created		How much the center of frequency energy moves across time	Take spectral centroid for each time period Now take the variance of these centroids
wavelet_energy_low_scale	Created		Measures the amount of energy at different wavelet scales, low scale = high frequencies. High scale = low frequencies	Use a wavelet transform
wavelet_energy_mid_scale	Created		Measures the amount of energy at different wavelet scales, low scale = high frequencies. High scale = low frequencies	Use a wavelet transform
wavelet_energy_high_scale	Created		Measures the amount of energy at different wavelet scales, low scale = high frequencies. High scale = low frequencies	Use a wavelet transform
regime_dominance	Created		Proportion of time spent in each regime, which regime dominates	Find regimes using previous team's code, then calc/find the proportion of the dominant regime.
rms_change_points	Created	Yes	How often the overall energy of the signal changes significantly over time	Find root mean square error, which is a measure of signal strength. Then divide signal into time windows, and find when a change in RMS is greater than 2 standard deviations, return how many changes are found.
skew_peak_magnitude	Created		The skewness of the peak magnitudes. Measures asymmetry of peak magnitudes distribution around its mean.	
kurtosis_peak_magnitude	Created		The kurtosis of the peak magnitudes. Measures how heavy the tails of the distribution are	
iqr_peak_magnitude	Created		The iqr of the peak magnitudes, higher iqr = more variation in peak size	
num_peak_mag_high_diff	Created		Pairwise feature: Number of peak magnitude differences greater than 2 standard deviations.	Find peak magnitudes of both accelerometers. Then compute pairwise differences. Now find std dev of differences, and find how many pairwise differences greater than 2 standard deviations there are.
peak_mag_correlation	Created		Pairwise feature: How similar the peak magnitudes are between the two accelerometers.	Find peak magnitudes of both accelerometers. Then find correlation between the two sets.
max_to_mean_peak_ratio	Created	No	How extreme the largest peak is compared with a typical peak. H	max_peak_magnitude divided by mean_peak_magnitude

Subcluster #	Feature Name	Previous or Created	Time Detrending	Intuitive Meaning	How it's calculated	Mean of Feature in Cluster	Signal Example 1	Signal Example 2	Signal Example 3
0 (Sporadic regime)	high_band_power	Created		The fraction of total signal energy in high frequency range	Total signal in high frequency	Low			
0 (Sporadic regime)	mean_spectral_entropy_time	Created		How spread out frequency is, averaged across time	Calc the spectrogram. Then normalize each time slice. Now calculate entropy at each time period. Take the mean of the entropy values across the different time periods	Low			
0 (Sporadic regime)	mean_to_max_peak	Created		How extreme the largest peak is compared with a typical peak. Higher value means one or a few peaks dominate.	max_peak_magnitude divided by mean_peak_magnitude	Low			
1 (Chaotic regime)	unused_peak_proportion	Previous		Fraction of peaks not found to be part of any detected regime	Count peaks not assigned to any regime, divide by total peaks (regime assignments decided by previous teams' code in num booleans)	High Unused Peak Proportion			
1 (Chaotic regime)	kurtosis	Created		Measures how heavy the tails of the distribution are. High kurtosis = occasional very high peaks	The fourth standardized central moment	High Kurtosis			
1 (Chaotic regime)	crest factor	Created		How extreme the peaks are relative to the overall "energy" of the signal. High crest factor = short, sharp spikes	Peak amplitude divided by root mean square of the signal	High Crest Factor			

Subcluster #	Feature Name	Previous or Created	Time Dendring	Intuitive Meaning	How it's calculated	Mean of Feature in Cluster		Signal Example (short)	Signal Example (medium)	Signal Example (long)
1 (Dense Rhythmic)	Spectral Entropy	Previous		How spread out the signal's energy is across frequencies (high = more spread out)	Calc power spectral density with Welch's method, normalize it, find Shannon entropy	Low Spectral Entropy	Example of low spectral entropy →			
1 (Dense Rhythmic)	Spectral Centroid	Previous		Where the average frequency of the signal lies (low frequencies = low centroid)	Weighted average of frequencies	Low Spectral Centroid	Example of high spectral centroid →			
1 (Dense Rhythmic)	Wavelet Energy (low scale)	Created		Measures the amount of energy at low scale = high frequencies. High wavelet energy means more of the signal is made up of small, rapid acoustic changes.	Use a wavelet transform	High Wavelet energy	Example of high wavelet energy →			
2 (Double Rhythmic)	Burstiness	Created		Measures how irregular the signal is. High burstiness = periods of rapid activity followed by slow periods	std deviation of inter peak arrivals divided by mean of inter peak arrivals	High Burstiness	Example of high burstiness →			
2 (Double Rhythmic)	num_boilings	Previous	Yes	Number of distinct patterns of boiling found	From previous team's code	High Number of Boiling patterns	Example of high num. Boilings →			
2 (Double Rhythmic)	regime_dominance	Created		Proportion of time spent in each regime, which regime dominates	Find regimes using previous team's code, then calc/find the proportion of the dominant regime.	Low Regime Dominance	Example of low regime dominance →			

Subcluster #	Feature Name	Previous or Created	Time Detrending	Intuitive Meaning	How it's calculated	Mean of Feature in Cluster		Signal Example (short)	Signal Example (medium)	Signal Example (long)
3 (Single Rhythmic)	Peaks_per_second	Previous	Yes	How many peaks appear per second on average	Total number of peaks divided by length of time	Low peaks per second	Example of low peaks per second →			
3 (Single Rhythmic)	regime_dominance	Created		Proportion of time spent in each regime, which regime dominates	Find regimes using previous team's code, then calc/find the proportion of the dominant regime.	High regime dominance	Example of high regime dominance →			
3 (Single Rhythmic)	num_boilings	Previous	Yes	Number of distinct patterns of boiling found	From previous team's code	Low Number of Boiling patterns	Example of low num. boilings →			

Note: we changed the names of Cluster 1 and 2 to fit the regime better!

REGIME LABELS FROM PREVIOUS GROUP:		
3 = Single Rhythmic Regime	1 = Dense Rhythmic Regime	2 = Double Rhythmic Regime
Regime 1: Single Rhythmic Signal for MATLAB 1:02 PM Thu, Nov 7, 2024 Run8 .csv Signal for MATLAB 4:57 PM Tue, Oct 1, 2024 Run11 .csv	Regime 2: Double Rhythmic Signal for MATLAB 1:00 PM Fri, Jun 28, 2024 Run8 .csv Signal for MATLAB 3:19 PM Wed, Sep 18, 2024 Run10 .csv	Regime 4: Rhythmic With Climax Signal for MATLAB 2:15 PM Thu, Oct 31, 2024 Run8 .csv